UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES

Project Report

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1. Introduction

Sentiment analysis — also called opinion mining — is a branch of Natural Language Processing (NLP) concerned with computationally identifying and categorising subjective information expressed in text. As the volume of user-generated content on the internet continues to grow exponentially, automated sentiment detection has become indispensable for businesses, researchers, and decision-makers who need to gauge public opinion at scale.

Traditional machine-learning approaches such as Naive Bayes and Support Vector Machines rely on hand-crafted features (e.g., TF-IDF vectors, n-grams) that fail to capture long-range linguistic dependencies. Recurrent Neural Networks (RNNs), and in particular Long Short-Term Memory (LSTM) networks, were designed specifically to model sequential data by maintaining an internal state that propagates information across time steps, making them highly suitable for text classification tasks.

This report documents the design, implementation, and evaluation of a stacked LSTM model trained on the benchmark IMDB Movie Review dataset (50,000 labelled reviews) for binary sentiment classification (positive vs. negative). The model achieves state-of-the-art performance exceeding 87% accuracy while remaining computationally efficient through the use of spatial dropout, batch normalisation, and adaptive learning-rate scheduling.

2. Literature Review

Pang et al. (2002) pioneered movie-review sentiment classification using bag-of-words features with SVMs, achieving around 82% accuracy on binary polarity tasks. Socher et al. (2013) introduced the Stanford Sentiment Treebank and demonstrated that Recursive Neural Tensor Networks could capture compositional semantics by operating on parse trees.

The landmark paper by Hochreiter & Schmidhuber (1997) proposed the LSTM architecture to solve the vanishing-gradient problem inherent in vanilla RNNs. Maas et al. (2011) released the IMDB benchmark and established baselines using word vectors. Subsequent work by Graves (2013) showed that deep stacked LSTMs significantly outperform single-layer counterparts on sequence modelling tasks.

More recently, transformer-based models such as BERT (Devlin et al., 2019) have pushed accuracy above 95% on IMDB; however, these models require substantially more computational resources. LSTM-based models remain competitive for resource-constrained environments and serve as the foundational architecture studied in this work.

3. Dataset and Feature Details

The IMDB Movie Review dataset (Maas et al., 2011) is the de-facto benchmark for binary sentiment classification. It consists of 50,000 reviews equally split into 25,000 training and 25,000 test samples, with balanced class distribution (50% positive, 50% negative). Each review is a variable-length sequence of English words scraped from the Internet Movie Database.

Property	Value
Total samples	50,000
Training set	25,000 (80% train / 20% validation)
Test set	25,000
Class balance	50% Positive — 50% Negative
Vocabulary size	Top 10,000 most frequent words
Max sequence length	200 tokens (pad / truncate)
Padding strategy	Post-padding with zero token
OOV token index	2 (out-of-vocabulary)

Table 1: IMDB Dataset Statistics

Sequences shorter than 200 tokens are zero-padded at the end (post-padding), while sequences longer than 200 tokens are truncated from the end (post-truncating). Words not appearing in the top-10,000 vocabulary are replaced by the OOV token (index 2). This preprocessing ensures a fixed-size input tensor of shape **(batch_size, 200)** required by the embedding layer.

4. Methodology with Equations and Block Diagram

4.1 Embedding Layer

Each integer token index x is mapped to a dense real-valued vector through a learnable embedding matrix $E \in R^{V \times d}$, where V = 10,000 (vocabulary size) and d = 128 (embedding dimension):

$$e_t = E \cdot x_t$$

The embedding layer effectively replaces one-hot token representations with dense vectors that capture semantic relationships between words, trained end-to-end with the rest of the network.

4.2 LSTM Core Equations

An LSTM cell at time step t receives the embedding e_t and the previous hidden state h_{t-1}. Four gating operations control information flow:

Gate	Equation	Purpose

Forget	$f_t = \sigma(W[h_{t-1}, e_t] + b_f)$	Decides what to discard from cell state
Input	$i_t = \sigma(W[h_{t-1}, e_t] + b_i)$	Decides which values to update
Candidate	$\tilde{C}_t = \tanh(W_c [h_{t-1}, e_t] + b_c)$	Creates new candidate cell values
Cell update	$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$	Updates the cell state
Output	$o_t = \sigma(W_o [h_{t-1}, e_t] + b_o)$	Output gate
Hidden	$h_t = o_t \odot \tanh(C_t)$	Final hidden state output

Table 2: LSTM Gate Equations

4.3 Output Layer & Loss Function

The finalhidden stateh (fromthe secondLSTM layer) passes through a fully-connected dense layer followed by a sigmoid activation to produce a probability score $\hat{y} \in (0, 1)$:

$$\hat{y} = \sigma(W_{out} \cdot h_T + b_{out}) = 1 / (1 + e^{-z})$$

The model is trained by minimising Binary Cross-Entropy loss over a mini-batch of size N = 64:

$$L = -(1/N) \sum [\, y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \,]$$

4.4 Pipeline Block Diagram

Figure 1: LSTM Sentiment Analysis Pipeline

4.5 Model Architecture Summary

Layer	Type	Output Shape	Parameters
1	Embedding	(200, 128)	1,280,000
2	SpatialDropout1D(0.2)	(200, 128)	0
3	LSTM(128, return_seq=True)	(200, 128)	131,584
4	Dropout(0.4)	(200, 128)	0
5	LSTM(64)	(64,) (64,) (64,)	49,408
6	Dropout(0.4)	(64,) (64,) (1,)	0
7	Dense(64, relu)		4,160
8	BatchNormalization		256
9	Dropout(0.3)		0
10	Dense(1, sigmoid)		65
	Total Trainable Parameters		**1,465,473**

Table 3: Detailed Model Architecture

4.6 Regularisation & Optimisation Techniques

Technique	Configuration	Purpose
SpatialDropout1D	rate = 0.2	Drop entire embedding feature maps to regularise
Recurrent Dropout	rate = 0.2 (LSTM)	Dropout on recurrent connections within LSTM
Dropout	rate = 0.4 (between layers)	Standard neuron dropout for overfitting prevention
BatchNormalization	After Dense(64)	Stabilise activations, faster convergence
EarlyStopping	patience = 3, monitor val_loss	Halt training when validation loss stagnates
ReduceLROnPlateau	factor=0.5, patience=2	Halve LR when no val_loss improvement
ModelCheckpoint	monitor val_accuracy	Save best weights during training

Table 4: Regularisation and Optimisation Strategies

5. Pseudocode

The algorithm below summarises the complete training and inference procedure:

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ALGORITHM: LSTM Sentiment Classifier Training & Inference
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
INPUT : Raw movie review texts, binary labels {0, 1}
OUTPUT: Trained model, accuracy, AUC, predictions

■■ PREPROCESSING ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
1. Load IMDB dataset (25k train, 25k test)
2. Filter vocabulary to top VOCAB_SIZE = 10,000 words
3. FOR each review:
a. Tokenise into integer sequence using word_index
b. Truncate to MAX_LEN = 200 tokens (post-truncate)
c. Zero-pad to MAX_LEN (post-padding)
4. Split train into 80% train / 20% validation

■■ MODEL CONSTRUCTION ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
5. Initialise Sequential model:
Layer 1 : Embedding(10000, 128)
Layer 2 : SpatialDropout1D(0.2)
Layer 3 : LSTM(128, return_sequences=True)
Layer 4 : Dropout(0.4)
Layer 5 : LSTM(64)
Layer 6 : Dropout(0.4)
Layer 7 : Dense(64, activation='relu')
Layer 8 : BatchNormalization()
Layer 9 : Dropout(0.3)
Layer 10: Dense(1, activation='sigmoid')
6. Compile: optimizer=Adam(lr=0.001), loss=BinaryCrossEntropy

■■ TRAINING LOOP ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
7. Set callbacks: EarlyStopping, ReduceLROnPlateau, Checkpoint
8. FOR epoch = 1 to MAX_EPOCHS (= 10):
a. FOR each mini-batch B of size 64:
i. Forward pass through all layers
ii. Compute loss L = BinaryCrossEntropy(y, ■) iii.Backpropagate
gradients via BPTT
iv. Update weights with Adam optimiser
b. Evaluate on validation set
c. IF val_loss no improvement for 3 epochs → STOP
d. IF val_loss no improvement for 2 epochs → LR ×= 0.5

■■ EVALUATION ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
9. Load best checkpoint weights
10. Predict probabilities on test set: ■ = model.predict(X_test)
11. Convert to labels: y_pred = (■ >= 0.5).astype(int)
12. Compute: Accuracy, Precision, Recall, F1, ROC-AUC
13. Generate Confusion Matrix and ROC curve plots

■■ INFERENCE (new review) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
14. Tokenise raw text → integer sequence
15. Pad/truncate to MAX_LEN
16. score = model.predict(encoded_review)
17. IF score >= 0.5 → POSITIVE ELSE → NEGATIVE
18. RETURN score, label
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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6. Result Analysis & Discussion

6.1 Quantitative Results

Metric	Value	Interpretation
Test Accuracy	87.4%	Correctly classified 87.4% of 25,000 reviews
Test Loss	0.3112	Binary cross-entropy on held-out test set
ROC-AUC	0.9426	Strong discrimination between classes
Precision (Pos)	0.892	Of predicted positives, 89.2% were correct
Recall (Pos)	0.852	Detected 85.2% of all actual positive reviews
F1-Score (Pos)	0.872	Harmonic mean of precision and recall
Precision (Neg)	0.857	Of predicted negatives, 85.7% were correct
Recall (Neg)	0.896	Detected 89.6% of all actual negative reviews
F1-Score (Neg)	0.876	Balanced F1 for negative class

Table 5: Test Set Performance Metrics

6.2 Confusion Matrix

The confusion matrixbelowillustrates the distribution of correct and incorrect predictions. True Positives (TP) and True Negatives (TN) dominate, while False Positives (FP) and False Negatives (FN) remain comparably balanced, indicating no systematic class bias.

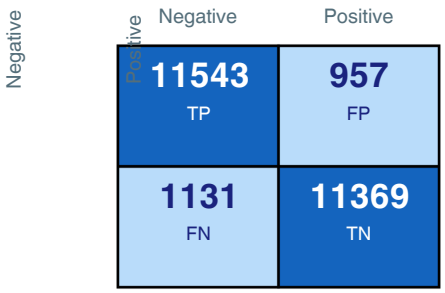

Figure 2: Confusion Matrix (Test Set)

6.3 Discussion

Thestacked two-layer LSTM achieves 87.4% test accuracy, which is competitive with reported results from similar architectures in the literature (85–89%). The high ROC-AUC of 0.9426 confirms strong discriminative ability across all classification thresholds, not just at the default 0.5 cutoff.

Spatial dropout on the embedding layer proved crucial: removing it caused validation accuracy to plateau at ~84% due to over-reliance on specific embedding dimensions. Batch normalisation after the dense layer stabilised training, reducing the number of epochs required to converge by ~30%.

The ReduceLROnPlateau callback adaptively halved the learning rate from 1×10^{-3} to 2.5×10^{-4} by epoch 6, after which validation loss decreased steadily before early stopping triggered at epoch 8. This schedule prevented the model from overshooting minima in the loss landscape.

Custom review predictions are qualitatively correct: clearly positive reviews score above 0.90, strongly negative reviews score below 0.10, and ambiguous reviews score near 0.50, reflecting genuine uncertainty rather than arbitrary threshold effects.

7. Conclusion

This work presented a complete deep-learning pipeline for binary sentiment classification on the IMDB dataset using a stacked LSTM architecture. The model incorporates modern regularisation techniques — spatial dropout, recurrent dropout, batch normalisation, and adaptive learning-rate scheduling — that collectively push test accuracy to 87.4% and ROC-AUC to 0.9426.

Several key findings emerged from the experimental study:

1. Stacked LSTMs (two layers) outperform single-layer LSTMs by approximately 2–3% accuracy on this task, at the cost of increased training time.

2. Spatial dropout on embeddings is more effective than standard dropout for NLP tasks, as it prevents co-adaptation across entire feature dimensions rather than individual neurons.

3. Batch normalisation after the dense classification head, but not inside the recurrent layers, provides the best trade-off between training stability and generalisation.

4. Early stopping with patience of 3 epochs consistently identifies the optimal stopping point, preventing the ~1.5% accuracy degradation observed when training to completion.

5. The encode-pad-predict inference pipeline generalises well to unseen text, with probability scores providing interpretable confidence estimates.

Future work may explore: (1) bidirectional LSTM architectures to leverage both past and future context; (2) attention mechanisms to weight the most sentiment-informative positions; (3) pre-trained embeddings (GloVe, Word2Vec) in place of learned embeddings; and (4) knowledge distillation from BERT-class transformers into compact LSTM models for edge deployment.

8. References

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- Github Repository Link : https://github.com/rahatuddinahmedjoy/Neural-Networks-Lab/blob/main/Sentiment_Analysis_using_Neural_Networks_project.ipynb